

FEASIBILITY STUDY

FloodSense: An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoor City

Presented to

Department of Computer Studies

Cavite State University — Bacoor City Campus

Submitted by:

Reymart V. Goc-ong

Martin Lorenz D. Juanites

Romar T. Pulao

May 2026

OVERVIEW

Flooding remains one of the most persistent and consequential natural hazards affecting Bacoor City, Cavite. Located in a low-lying coastal area within Cavite province, Bacoor City is regularly exposed to flooding driven by monsoon rainfall, storm surges, and inadequate drainage infrastructure. As supported by the findings of the primary data gathering phase of this study, the impact of flooding on residents is not only physical but also informational — residents consistently lack timely, accurate, and location-specific flood susceptibility information that would allow them to make informed decisions before and during flood events.

A structured survey administered to 50 residents across 12 flood-prone barangays in Bacoor City revealed significant gaps in the current state of flood information dissemination. Of all respondents, 92% reported experiencing flooding in their area, with 54% experiencing it almost every rainy season and 58% reporting severe flooding. Despite the frequency and severity of flooding, only 20% of respondents considered themselves well-informed about flood-prone areas in Bacoor City — with the remaining 80% falling under “Somewhat Aware,” “Not Sure,” or “Not Aware at All.” The three Likert-scale items on flood information quality (Q9–Q11) yielded mean scores of 2.62, 2.70, and 2.72 respectively, indicating near-dissatisfied perceptions of current information quality, timeliness, and clarity. Most critically, 64% of respondents — 32 out of 50 — reported having made a wrong decision due to insufficient flood information, including taking flooded roads, failing to evacuate in time, and misjudging flood conditions.

Thematic analysis of open-ended survey responses further confirmed the nature of this gap. The two most prevalent themes from Q12 — “Biggest Problem in Getting Flood Information” — were Late or Delayed Information (26%) and Lack of Information or No Warnings at All (20%), together accounting for 38% of all responses. Respondents also cited unclear or inaccurate information (12%) and lack of location-specific flood susceptibility information (6%) as recurring barriers. The survey findings were subsequently validated through a Key Informant Interview with a staff member of the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), who confirmed that the office’s current flood information dissemination relies primarily on social media, email, and barangay coordination — and acknowledged that resident engagement with official warning channels remains low due to poverty, work constraints, and low attendance at flood orientations.

The BDRRMO interview further revealed that while the office maintains historical flood data and GIS-based exposure maps per barangay, these datasets are not accessible to the general public in a usable or interactive format. The interviewee confirmed that the office’s data holdings include barangay boundary maps, historical flood records (affected areas and flood events), and field-assessed exposure data — and expressed openness to data sharing for research and academic purposes. Critically, the BDRRMO officer affirmed that a mobile-based tool providing location-specific flood susceptibility information would serve a genuine community need — particularly for commuters, travelers, and residents in **highly susceptible barangays** who require actionable flood awareness support for area-avoidance and evacuation decisions.

FloodSense is a proposed mobile application that addresses these identified gaps through the integration of machine learning-based flood susceptibility classification and augmented reality visualization. Using a supervised ML classification model trained on environmental features — including historical flood records, elevation data, drainage proximity, soil type, and rainfall parameters — the system classifies **flood susceptibility** across Bacoor City’s barangays into four levels: Low, Moderate, High, and Very High. These classifications are rendered through an augmented reality ground-plane overlay, allowing users to visualize susceptibility levels by pointing their phone camera at any flat surface. Rainfall input is simplified through a PAGASA-aligned dropdown menu using plain-language intensity presets (Light, Moderate, Heavy,

Intense/Torrential), ensuring usability for non-technical community users. The system targets all residents, commuters, and travelers navigating Bacoor City — providing a decision support tool that bridges the gap between institutional flood data and actionable community-level flood awareness.

KEY FEATURES

Core Features

1. Augmented Reality Flood Susceptibility Map

Renders a color-coded flood susceptibility map (Low/Moderate/High/Very High) per barangay through the phone camera using Vuforia ground-plane markerless tracking. The AR overlay is **anchored to a detected flat surface** — not GPS-linked to the user's outdoor position. A surface detection indicator (green = valid, red = invalid) ensures accurate rendering before the map is displayed. The map shows a Legend overlay only; title, date, and data source are accessible through an in-app info panel to avoid AR clutter.

- Color-coded susceptibility levels: Low (green), Moderate (yellow), High (orange), Very High (red)
- Markerless ground-plane tracking via Vuforia — works on any flat horizontal surface (table, floor)
- Surface detection indicator — green/red; map inputs locked until surface confirmed valid
- Legend visible in AR view; full metadata (STANDL elements) in info panel
- Disclaimer accessible from AR view: susceptibility levels indicate physical flood likelihood, not a real-time guarantee

2. PAGASA-Aligned Rainfall Input

Users select rainfall scenario and duration from dropdown menus using plain-language labels tied to PAGASA's public rainfall advisory classification. The app internally maps these selections to millimeter-per-hour values used by the ML model — removing the need for technical numeric input from non-expert users. This design was directly validated by the BDRRMO key informant.

- Rainfall intensity presets: Light (0–7.5 mm/hr), Moderate (7.5–15 mm/hr), Heavy (15–30 mm/hr), Intense/Torrential (>30 mm/hr)
- Duration dropdown: 1 hr, 3 hrs, 6 hrs, 12 hrs
- Rain intensity icons alongside text labels (optional)
- Reference: PAGASA Rainfall Advisory System

3. Machine Learning Flood Susceptibility Classification

A supervised classification model trained on environmental datasets classifies flood susceptibility per barangay based on user-selected rainfall parameters combined with pre-loaded spatial features. The model outputs one of four susceptibility levels per zone, which directly drives the AR map color coding. All spatial features (elevation, soil type, drainage proximity) are pre-identified by the research team from institutional maps — not collected from community surveys.

- Classification output: Low / Moderate / High / Very High

- Features: elevation, drainage proximity, soil type (researcher-identified from NAMRIA maps), historical flood records, rainfall intensity and duration
- Data sources: BDRRMO historical records, NAMRIA elevation data, CPDO land use maps, DOST-ASTI / PAGASA rainfall data
- Model: Random Forest or XGBoost (scikit-learn, Python)

4. User Account and Preference System

Users create an account to access personalized features including saved rainfall preferences. Firebase Authentication handles user management with secure credential storage.

- Email/password account creation and login
- Saved rainfall preference parameters (intensity and duration)
- User profile and settings management

5. Onboarding and Disclaimer Screen

A first-launch onboarding screen informs users of the app's purpose, limitations, and proper usage — ensuring informed use and reducing the risk of misinterpreting susceptibility outputs as real-time flood guarantees.

- App purpose and scope explanation
- Disclaimer: 'This app models rainfall-induced flood susceptibility. It is not a real-time flood sensor. Always follow official warnings from BDRRMO and PAGASA.'
- Flat surface instruction for AR usage

Additional Features (Stretch Goals — Phase 4 Only)

Scope Note

The following features are deferred to Phase 4 and will only be developed if Phases 1–3 are completed ahead of schedule. They are NOT core thesis objectives and will NOT be presented as primary deliverables at the title defense.

6. Flood Information Hub

A static in-app resource section providing barangay-level flood history, evacuation route references, and BDRRMO contact information for official guidance during flood events.

7. Offline Mode

Core susceptibility classification and AR map rendering function without internet connectivity using locally cached model and barangay data. Auto-syncs updates when connection is restored.

I. TECHNICAL FEASIBILITY

This section evaluates whether FloodSense can be built using currently available technologies, tools, and frameworks within the capability of an undergraduate Computer Science team. Each component is mapped to a specific, free or low-cost technology with a documented implementation path.

1.1 Proposed Technology Stack

Stack Decision	Flutter and React Native have been removed from the stack. All application screens (Login, Preferences, Results Summary, AR View) are built within Unity using the uGUI system. This eliminates the need for a separate mobile framework and a Unity-to-mobile handoff layer, reducing complexity and maintaining a single codebase.
-----------------------	--

Layer	Technology	Cost	Justification
Mobile App (all screens)	Unity + uGUI (Canvas, Input Fields, Dropdowns)	Free (Personal License)	Unified development environment for both UI screens and AR; eliminates Flutter/React Native handoff
AR Engine	Vuforia SDK (Ground Plane)	Free tier	Markerless ground-plane tracking on flat surfaces; no GPS anchoring required
Backend / Database	Firebase (Firestore, Auth)	Free tier	Real-time sync, built-in authentication, cloud storage
ML Model	Python — scikit-learn / XGBoost	Free	Flood susceptibility classification; trained on open and BDRRMO datasets
ML Serving	Firebase Functions or FastAPI (lightweight)	Free	Serves model predictions to Unity app via API call
Elevation / Spatial Data	NAMRIA, DOST-ASTI, SRTM	Free (open access)	Topographic and land use features for ML model training
Historical Flood Data	BDRRMO (via data-sharing agreement)	Free (institutional)	Ground-truth flood occurrence data per barangay
Rainfall Reference	PAGASA Advisory Classification	Free (public)	Basis for user-facing rainfall intensity presets
UI/UX Prototyping	Figma	Free tier	Mobile interface design and prototyping prior to Unity development

1.2 Feature-Level Technical Assessment

Feature	Implementation Approach	Complexity	Est. Time
AR Flood Susceptibility Map (Vuforia Ground Plane)	Unity + Vuforia ground-plane detection; flat-surface anchored	High	4 weeks

Feature	Implementation Approach	Complexity	Est. Time
	zone overlay; color-coded susceptibility rendering		
PAGASA Rainfall Dropdown + ML Inference	Unity uGUI Dropdown mapped to mm/hr values; API call to ML model; returns susceptibility level per barangay	Medium	2 weeks
ML Classification Model (Python)	scikit-learn/XGBoost trained on elevation, soil, drainage, historical flood, rainfall features; exported model served via API	High	4 weeks
Results Summary Screen	Unity uGUI ScrollView / List of barangay names + color badge; fed directly from ML API response stored in Firestore	Low	1 week
User Account + Auth (Firebase)	Firebase Authentication for email/password login; Firestore for user preferences	Low	1 week
Surface Detection Indicator	Vuforia ground plane status callback; tint material green/red based on surface validity; lock input until green	Low	1 week
Onboarding + Disclaimer Screen	Unity first-launch scene with instructions and disclaimer; dismissable with user acknowledgement	Low	0.5 week
Flood Information Hub (stretch)	Static Firestore-backed content screen; barangay flood history, evacuation info, BDRRMO contacts	Low	1 week
Offline Mode (stretch)	Locally cached ML model and barangay data; Firebase offline persistence for user data	Medium	2 weeks

1.3 Technical Challenges and Mitigations

AR Rendering Performance on Mid-Range Devices

Vuforia ground-plane AR and zone overlay rendering may cause performance issues on entry-level Android devices.

Mitigation: Optimize barangay zone geometry by simplifying polygon vertices. Implement level-of-detail rendering. Set minimum device specification in app description.

ML Model Accuracy with Limited Local Data

Historical flood data from BDRRMO covers affected areas but may lack granularity needed for high-accuracy susceptibility classification.

Mitigation: Supplement BDRRMO records with DOST-ASTI Project NOAH hazard maps and PAGASA historical rainfall data. Apply k-fold cross-validation during training. Report model performance metrics (accuracy, precision, recall, F1) transparently in the study.

User Input Accuracy for Rainfall Selection

Users may not correctly identify rainfall intensity without reference, leading to inaccurate model queries.

Mitigation: Use PAGASA-aligned plain-language labels (Light, Moderate, Heavy, Intense) with optional rain intensity icons. Add brief instruction: 'Select the rainfall level that best matches current or expected conditions.' Directly validated by the BDRRMO key informant.

BDRRMO Data Format and Availability

BDRRMO historical flood data may be in paper records or inconsistent digital formats not immediately usable for ML feature engineering.

Mitigation: Request data in Excel or GIS format through a formal data-sharing request. If incomplete, use DOST-ASTI and PAGASA open datasets as primary training sources and treat BDRRMO data as supplementary validation input.

Soil Type Feature in Urbanized Areas

As noted by the BDRRMO interviewee, most of Bacoor's land surface is paved or built-up, making raw soil type a potentially weak feature in urbanized barangays.

Mitigation: Source soil type from NAMRIA land use classification maps — not from user input or community surveys. For paved/built-up areas, use impervious surface classification as a proxy feature. Researchers identify all spatial features independently from institutional maps.

Flat Surface Detection Edge Cases

Users may point the camera at non-flat surfaces (grass, fabric, uneven floors), preventing ground-plane detection.

Mitigation: Vuforia's Ground Plane automatically detects surface validity. Implement green/red surface indicator and a persistent instruction message: 'Point camera at a flat table or floor.' Block input form from appearing until surface is confirmed green.

II. OPERATIONAL FEASIBILITY

This section evaluates whether FloodSense can be realistically developed, deployed, and used within the constraints of the thesis timeline, team size, target users, and available institutional resources.

2.1 Team Composition and Role Distribution

Member	Primary Role	Responsibilities
Member 1	Mobile / AR Developer	Unity uGUI screen development (Login, Preferences, Results Summary, AR View), Unity scene navigation, user account UI, onboarding flow, Vuforia AR integration, surface detection UI
Member 2	ML Engineer / Backend	Python ML model development (feature engineering, training, evaluation), Firebase setup, ML model serving via API, Firestore integration
Member 3	Data / Research Lead	Spatial data collection and preprocessing (NAMRIA, DOST-ASTI, BDRRMO), barangay zone preparation, usability testing coordination, thesis documentation

2.2 Recommended Development Timeline (10 Months)

Phase	Duration	Deliverables
Phase 1 — Foundation	Months 1–2	Project setup, Firebase configuration, user authentication, basic Unity UI screens, BDRRMO and CPDO data request and collection, barangay zone data preparation
Phase 2 — ML Model Development	Months 3–4	Feature engineering from NAMRIA, DOST-ASTI, BDRRMO datasets; model training and evaluation (Random Forest / XGBoost); model export and API deployment
Phase 3 — AR Integration	Months 5–6	Vuforia Unity ground-plane AR setup; flat-surface barangay zone overlay; color-coded susceptibility rendering; surface detection indicator; legend overlay
Phase 4 — App Integration & Features	Months 7–8	PAGASA rainfall dropdown integration; ML API connection; user account and preferences; Results Summary screen; onboarding/disclaimer screen; Flood Information Hub and offline mode if time allows
Phase 5 — Testing & Defense	Months 9–10	Usability testing with Bacoar residents; ML model evaluation reporting; bug fixes; BDRRMO review of output; thesis documentation; final defense preparation

2.3 Target Users and Adoption Feasibility

Scope Boundary	All user tiers interact with the system at barangay level only. The app answers: ‘Which barangays in Bacoar City are at Low / Moderate / High / Very High flood susceptibility given this rainfall scenario?’ It does not provide street-level routing, real-time flood tracking, or navigation guidance.
-----------------------	---

User Tier	Description	Adoption Strategy
Tier 1 — Bacoar City Residents (flood-prone barangays)	Residents of historically susceptible barangays (Zapote, Mambog, Talaba, Niog) who experience flooding regularly and need location-specific susceptibility information for evacuation and safety decisions	Pilot distribution through BDRRMO community outreach and barangay-level orientation; direct recruitment from survey respondents
Tier 2 — Commuters and Travelers	Individuals traveling to or through Bacoar City who need barangay-level flood susceptibility information to make informed decisions about which areas of the city to avoid during heavy rainfall scenarios — not for street-level routing, but for zone-level risk orientation. Validated by BDRRMO interview as a primary use case.	App store distribution; social media promotion through BDRRMO and barangay official channels
Tier 3 — General Bacoar Residents	Residents outside the highest-susceptibility zones who use the app for general flood awareness and community preparedness planning	App store distribution; awareness through city-level flood information campaigns

2.4 Financial Feasibility

FloodSense is designed to operate at zero licensing cost during the thesis development and pilot testing phases. All tools listed operate within free tiers sufficient for a thesis-scale pilot (20–50 participants).

Tool / Service	Purpose	Cost
Unity + Vuforia SDK	AR development, ground-plane tracking, and all UI screens (uGUI)	Free (Personal License)
Firebase Spark Plan	Auth, Firestore database, Cloud Functions	Free tier
Python / scikit-learn / XGBoost	ML model training and evaluation	Free
NAMRIA /CPDO Topographic Data	Elevation and soil type features for ML model	Free (open access)
DOST-ASTI / Project NOAH / BDRRMO	Flood hazard maps and rainfall data	Free (open access)
PAGASA Rainfall Advisory Data	Rainfall intensity classification reference	Free (public)
BDRRMO Historical Flood Data	Ground-truth barangay flood records	Free (data-sharing agreement)
Figma	UI/UX prototyping	Free tier

2.5 Operational Risks and Mitigations

BDRRMO Data Access Delay

BDRRMO may require additional time to process a formal data-sharing request, potentially delaying ML model development.

Mitigation: Initiate formal data request immediately after title defense approval. Begin ML model development in parallel using DOST-ASTI and PAGASA open datasets as primary sources. Treat BDRRMO data as supplementary validation input to avoid single-source dependency.

CPDO Spatial Data Inaccessibility

CPDO may not have digital GIS files readily available, or may store data in formats incompatible with the research pipeline.

Mitigation: Request data in Excel, shapefile, or any available digital format. Use NAMRIA national topographic data and PhilGIS land use classification as fallback sources. Document data source substitutions in methodology.

Low Pilot User Adoption

Community residents, particularly elderly or low-literacy users, may find the AR interface unfamiliar or difficult to use.

Mitigation: Conduct usability testing with at least 20 residents from the original survey pool. Simplify onboarding with visual instructions. Leverage BDRRMO community outreach as a distribution channel for pilot recruitment.

Scope Creep — AR + ML Dual Complexity

Developing both an AR system (Unity/Vuforia) and an ML pipeline (Python) in parallel is technically demanding for a thesis team.

Mitigation: Assign dedicated roles — AR/UI development and ML development handled by separate team members. Phase timeline separates ML completion (Phase 2) from AR integration (Phase 3) to prevent bottlenecks. Offline mode and Flood Information Hub are stretch goals deferred to Phase 4 only if earlier phases are completed ahead of schedule.

Data Privacy Compliance

User account data is subject to Philippine Republic Act 10173 (Data Privacy Act).

Mitigation: Firebase Security Rules enforce per-user data isolation. Informed consent obtained from all pilot participants. No personal location data is collected, stored, or transmitted. The AR overlay is anchored to a detected flat surface — the user's GPS coordinates are not used or stored by the application.

Sustainability Post-Thesis

Without a formal data-sharing agreement, the app's ML model may become outdated as new flood events occur.

Mitigation: Frame sustainability as future work in the thesis defense. Proposed sustainability pathway: establish a formal MOU with BDRRMO for periodic model retraining using updated historical flood records, and explore integration with BDRRMO's existing community information dissemination channels.

Interview-Identified Gap: User Input Accuracy

As raised by the BDRRMO key informant, users cannot accurately estimate rainfall in millimeters, making free-text or numeric input unreliable.

Mitigation: Fully addressed through the PAGASA-aligned dropdown design — users select plain-language intensity levels (Light, Moderate, Heavy, Intense) tied to standardized mm/hr ranges. The app handles all technical conversion internally. This design decision was directly validated by the BDRRMO interview.

Interview-Identified Gap: Output Clarity for Non-Expert Users

The BDRRMO interviewee raised that users unfamiliar with map conventions may misread susceptibility levels as absolute flood predictions rather than classification-based outputs.

Mitigation: Include a brief legend explanation accessible from the AR view. Add disclaimer on first launch: 'Susceptibility levels indicate the likelihood of flooding based on environmental conditions — not a guarantee of actual flooding.' Brief instructional tooltip accessible during AR use.

2.6 Feasibility Summary

Dimension	Assessment	Verdict
Technical	All features implementable with free, well-documented tools; ML susceptibility classification model trained on open and BDRRMO datasets using scikit-learn/XGBoost; AR rendering via Vuforia ground-plane tracking on flat surfaces — no GPS anchoring required; all screens built in Unity uGUI	FEASIBLE
Financial	Zero licensing cost during thesis period; all tools, APIs, and frameworks operate within free tiers sufficient for thesis-scale pilot testing (20–50 participants)	FEASIBLE
Team Capacity	Three-person team with clearly separated AR/UI, ML, and data roles; 10-month phased timeline isolates ML and AR development to prevent bottlenecks	FEASIBLE
User Adoption	Pilot testing achievable through BDRRMO community outreach and direct recruitment from survey respondents; PAGASA-aligned plain-language input removes technical barrier for non-expert users	FEASIBLE
Institutional Support	BDRRMO key informant confirmed openness to data sharing and collaboration; barangay boundary maps and	FEASIBLE

Dimension	Assessment	Verdict
	historical flood records are available for research use upon formal request	
Scope	Tidal flooding and flash floods deferred to future work; vulnerability-adjusted risk scoring deferred to future work; Flood Information Hub and offline mode are stretch goals only; remaining core features are well-scoped for thesis-level research within 10 months	FEASIBLE
Academic Contribution	Novel integration of AR visualization and ML-based flood susceptibility classification for a Philippine LGU context; PAGASA-aligned community input design; BDRRMO-validated problem and concept; correct use of susceptibility (not risk) framing supported by environmental feature set	STRONG